eXARA: Problem Statement & Implementation Plan

Problem Statement

Globally, about 7,000 languages are spoken, yet only about 20 receive substantial attention in NLP research (Joshi et al., 2020). Africa, home to over 2,000 languages, has seen a staggering 88% of its languages either severely underrepresented or completely ignored in computational linguistics (Issaka et al., 2025). Approximately 814 African languages are currently in danger of extinction, with Nigeria alone having 171 languages facing the most severe threats (Issaka et al., 2025). These are not marginal dialects spoken by tiny populations because languages like Igbo, Yoruba, Hausa, Twi, Efik, and Wolof are spoken by tens of millions of people each. Their near-total absence from modern AI systems is a reflection of the data infrastructure that has never been built for them (Issaka et al., 2025).

The reason that infrastructure does not exist is well understood. AI language systems are trained on enormous volumes of labelled, structured, validated text and speech data. For English, Mandarin, and Spanish, that data has accumulated over decades through digitized publications, transcribed broadcasts, annotated corpora, and the natural residue of digital communication at scale. Nigerian languages like Igbo and Yoruba, despite their widespread use, are classified as low-resource languages in NLP primarily due to the lack of digital data (Inuwa-Dutse, 2025). These languages collectively have fewer than 200,000 Wikipedia articles, which is a significant bottleneck for training NLP models (Inuwa-Dutse, 2025). The problem is therefore infrastructural, as there is no scalable mechanism for generating, collecting, and validating African language data at the volume the field requires (9jaLingo, 2026).

Researchers have not ignored this challenge. Since 2019, a number of community-driven and institutional initiatives have made meaningful contributions to African language NLP (9jaLingo, 2026). The Masakhane project established a participatory approach to machine translation across more than 30 African languages, creating parallel datasets through distributed community annotation, and demonstrated the potential of collaborative resource creation (Nekoto et al., 2020). Subsequent efforts produced standardised benchmarks including MasakhaNER 2.0 for named entity recognition across 20 languages, MasakhaPOS for part-of-speech tagging, and MAFAND-MT for news-domain machine translation. In speech recognition specifically, AfriSpeech-200 introduced a Pan-African accented English speech dataset crowd-sourced from 2,463 African speakers, covering 200 hours across clinical and general domains (Olatunji et al., 2023). Mozilla Common Voice has brought community-sourced recordings to languages including Hausa, Kinyarwanda, and Twi. Meta's multilingual speech initiative has extended ASR capability to hundreds of underserved languages (Pratap et al., 2023).

These efforts are genuinely valuable, but they share limitations that prevent them from closing the data gap at the scale required. Mozilla Common Voice recordings follow scripted prompts and primarily capture read speech, limiting applicability to spontaneous conversational settings. Masakhane's continental scope necessarily limits the depth of coverage for any individual language, and its focus on machine translation means it does not document critical metadata needed for resource selection; such as dataset sizes, licensing terms, domain coverage, or preprocessing requirements (Adjovi et al., 2026). Meta's approach, while representing a positive development, still results in less than 50 hours of data for many African languages (some with less than 10 hours) a very small fraction of the data on which supervised fine-tuning models are trained (Pratap et al., 2023). The fundamental constraint across all of these initiatives is that they rely on small, motivated cohorts of volunteers or paid annotators working in structured settings, and they produce data at a rate that does not match the scale of what is needed.

Within this broader data crisis sits a problem that is more granular and, in many ways, more urgent: the fragmentation and erosion of African language dialects and Igbo is an instructive example. The Igbo language encompasses several dialects distinguished by accents and orthography, including Owerri, Awgu, Ngwa, Umuahia, Nnewi, Onitsha, Awka, Abiriba, Arochukwu, Nsukka, Mbaise, and others. The diverse range of spoken dialects has posed significant challenges in establishing a standardised orthography, resulting in the current Onwu orthography of 1962 as a compromise (Igbo BC / IDAOBC, 2024). The speech varieties of Igbo include over twenty documented dialects (Afikpo, Aniocha, Azumini, Ngwa, Mbaise, Nnewi, Nsukka, Onitsha, Owerri, Umuahia, and others) each with distinct phonological, lexical, and grammatical features (Nganga & Achebe, 2020). A speaker from Abiriba and a speaker from Nnewi may both identify as Igbo speakers and yet struggle to fully understand each other in natural conversation, particularly on topics where local vocabulary diverges significantly (Dom-Anyanwu et al., 2023). These unique characteristics, including tonal variations and complex morphological structures, complicate the adaptation of NLP tools and mean that inconsistent diacritic usage negatively affects text readability and performance in downstream tasks, especially in text-to-speech applications (WorldSchoolbooks, 2024). As urbanisation draws younger generations toward cities where a flattened, simplified Igbo competes with English and Nigerian Pidgin, these dialects face active threat of disappearance. There is currently no digital system that captures, preserves, or facilitates communication across Igbo dialects, and no dataset that would make building one possible.

This is the problem eXARA is designed to solve, simultaneously at the level of data infrastructure and at the level of lived linguistic experience. eXARA is a gamified mobile application that turns everyday African language speakers into active contributors to the datasets that AI researchers need. In doing so, it makes the act of contribution feel less like research and more like play. A player selects their native language and dialect, and is then presented with a stimulus: not merely a static image, but potentially a short video clip, a recorded action, a gesture, or a real-world event unfolding on screen. They record a voice description of what they see in their chosen language and dialect. The breadth of stimulus types is deliberate, for example, a cup on a table and a woman grinding pepper in a mortar require fundamentally different descriptive vocabularies, and it is precisely that range of everyday, spontaneous, contextually grounded speech that existing African language datasets lack almost entirely.

The more sophisticated and data-rich mode is the multiplayer phase, where between two and eight players engage simultaneously in the same session. Here, one player takes the role of selector; choosing the image, video, or action that will serve as the shared stimulus, while the remaining players, who may speak the same dialect or different dialects of the same language, each independently produce their own spoken description of what they observe. The group then engages in a collective validation process: players review one another's descriptions, vote on correctness, surface disagreements, and collaboratively approve or refine responses that are partially accurate. Where a description is wrong, the group corrects it. Where a description is right, the group confirms it. The outcome of each session is a cluster of validated, cross-verified, dialect-annotated descriptions of the same stimulus.

Critically, the system is designed to learn from itself in a way that compounds in accuracy over time. Every description that clears the validation threshold whether through the dedicated human validator

(Studio) pipeline or through multiplayer peer consensus, is immediately enrolled as reference data against which future submissions are evaluated. When a new player from the same dialect subsequently describes the same or a similar stimulus, their response is assessed not only by the AI's semantic judgment but against the growing corpus of prior approved responses from real, verified speakers of that exact dialect. The validation standard therefore becomes progressively more dialect-accurate and culturally grounded with every session played. All submissions therefore, benefit from the accumulated voice of the community that preceded them. This self-reinforcing architecture means eXARA's data quality does not plateau at scale but improves with it. The more speakers contribute across more dialects, the more precise the validation becomes, and the more useful the resulting corpus is for training the dialect-aware speech and translation systems that are eXARA's ultimate scientific output.

But the data collection layer is only the foundation. The long-term vision of eXARA is to use that data to build something that does not yet exist anywhere in African technology: a real-time cross-dialect and cross-language translation system designed specifically for African linguistic contexts. The goal is a future in which a person speaking Abiriba dialect can have a natural conversation with a person speaking Nnewi dialect, with each hearing the other's words rendered in their own familiar speech, mediated invisibly by a system trained on eXARA's collected data. This same architecture then scales across language pairs: Igbo and Yoruba, Hausa and Igbo, Efik and Yoruba, and beyond. The data collection game and the translation infrastructure are not separate products but sequential layers of the same system.

Like eXARA, gamified data collection has been explored in other contexts, and crowdsourced speech corpora exist for some African languages. What does not exist is a system that collects dialect-level spontaneous speech data through an engagement mechanism designed for smartphone users in African markets, validates that data in real time without requiring prior ground truth, and feeds it directly into a pipeline aimed at cross-dialect intelligibility. The specific problem of intra-language dialect bridging has not been addressed in any published African NLP system. Tone marking in Yoruba and Igbo remains inconsistently handled in most existing datasets, code-switching corpora are rare, and languages like Efik, Tiv, Igala, and hundreds of others have almost no labelled data. eXARA enters a field where the problem is well-documented, the need is urgent, and the specific approach being proposed (gamified, dialect-granular, real-time validated, spontaneous speech collection oriented toward cross-dialect translation) has no direct precedent.

The societal impact of this solution operates at several levels. For individual users, eXARA offers the first digital communication tool that meets them in their own dialect rather than asking them to approximate a standardized form. For language communities, it creates a participatory, technology-forward mechanism for preserving dialects that are under genuine threat of disappearance within a generation. For AI researchers and developers, it generates the raw material that is the prerequisite for building any serious African language AI system. And for the broader African technology ecosystem, it establishes a model for community-driven, incentivized language data infrastructure that does not depend on foreign institutions, volunteer fatigue, or academic grant cycles.

The applications of this solution extend well beyond the game itself. In healthcare, a cross-dialect translation layer could allow a doctor from Onitsha to accurately communicate with a patient from Abiriba without either resorting to English. In education, teachers could deliver instruction in a student's home dialect while curriculum materials remain in a standardised form. In financial services, voice-based banking and advisory tools could serve rural populations who are fluent in their local dialect but not in

English or Standard Igbo. In governance, public information (health directives, civic notices, legal rights) could be communicated in the dialect a citizen actually speaks. In each of these sectors, the current absence of dialect-aware AI is not a minor inconvenience. It is a structural barrier to access, equity, and participation in modern society.

On the question of validation; how eXARA determines that a dialect description is true and correct, the system takes a layered approach precisely because ground truth does not yet exist. The first layer is structural: the system verifies basic file integrity, minimum audio duration, and metadata to ensure a genuine recording attempt was made. The second layer is human-in-the-loop expert validation: certified dialect speakers (Validators) review the audio against the image, manually transcribe the pure dialect text from scratch, and score the relevance, effort, and linguistic consistency. The third layer, which emerges as the dataset grows, is consensus-based: when multiple independent validators correct and approve descriptions of the same image that converge acoustically and semantically, the convergence itself becomes the validation signal. No single submission is treated as ground truth. Truth is established through agreement across independent experts. This is precisely the mechanism through which Common Voice and AfriSpeech established quality standards in their early phases but it is being applied here at dialect granularity for the first time.

On the question of motivation; why people will use eXARA and how the economy works, the design draws on a dual-tiered gamification system. In the first tier, everyday users act as Data Contributors. They record descriptions of images in their dialect and earn Experience Points (XP) and leaderboard status. This gamified progression unlocks more complex image sets but does not yield financial reward. However, once a contributor reaches a strict quality threshold, specifically accumulating 500 XP from consistently approved recordings, they are promoted to the second tier: Data Validators. It is these certified experts who earn actual eXARA Coins by manually reviewing, transcribing, and correcting the submissions of tier-one contributors. These Coins translate to real-world value through a reward mechanism: airtime, mobile data, or cash equivalent. The communities whose languages eXARA needs most are often communities where economic incentives matter enormously. Tangible, immediate, and fair compensation for the critical, high-friction work of dialect validation ensures eXARA remains viable at scale where every previous volunteer-based initiative has plateaued.

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